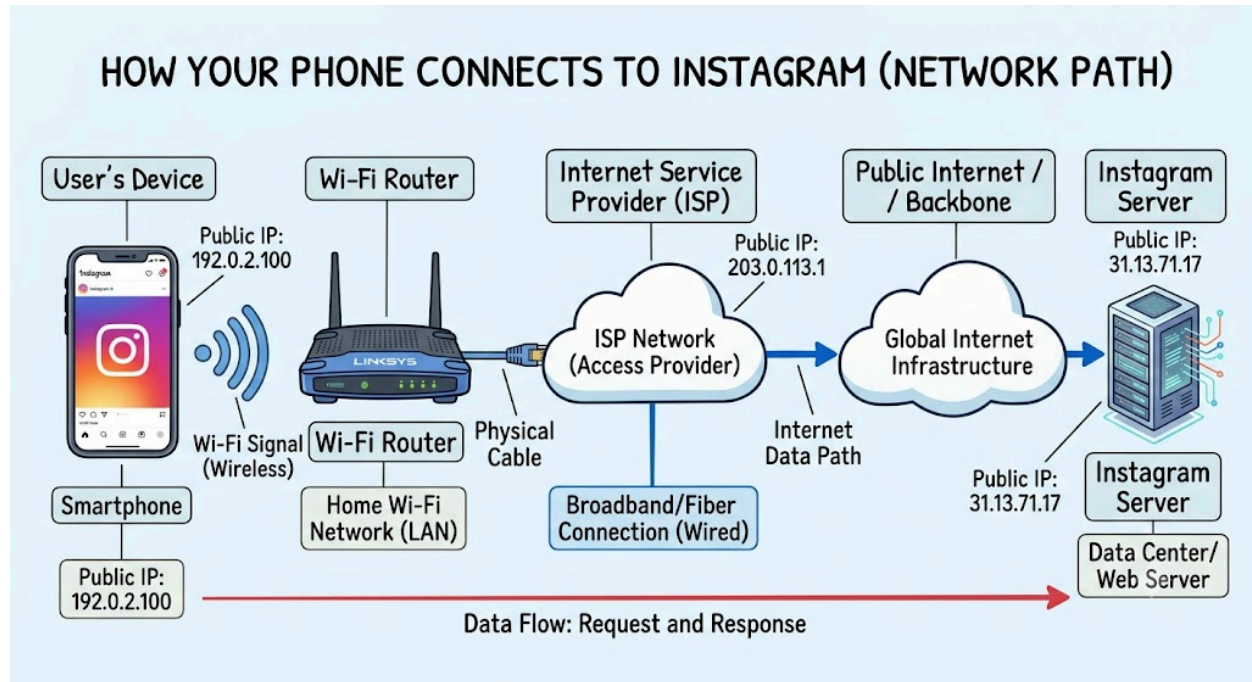


INTRODUCTION TO NETWORKS

Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.

ANS:



List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).

ANS:

Smartwatch: PAN (Bluetooth to phone)

Laptop: LAN / WLAN (Wi-Fi router)

Smart TV: LAN / WLAN (Wi-Fi/Ethernet)

Smartphone (Mobile Data): WAN (4G/5G cellular)

Wi-Fi Router: LAN (Internal) / **WAN** (ISP connection)

Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.

ANS:

Internet: A global, public network accessible to anyone (e.g., using campus Wi-Fi to browse Wikipedia or watch YouTube).

Intranet: A private, restricted network accessible only to authorized users (e.g., using the same Wi-Fi to access your college student portal/LMS for grades and attendance, which cannot be reached from outside).

Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.

Hint: Mention the devices and networks involved, and what happens when you hit 'search'.

ANS:

When you search for a restaurant on **Zomato**:

1. Tapping **Search** sends an HTTPS data packet with your query from your smartphone over a **PAN/WLAN** (Wi-Fi or cellular connection).
2. The packet reaches your **Wi-Fi Router**, where **NAT** translates your device's private IP to a public IP address.
3. Your **ISP** routes the packet across the **WAN** to Zomato's cloud infrastructure.
4. Zomato's **Firewall/WAF** inspects the traffic, forwards it to their application servers and database, and returns the matching restaurant results back along the same path to your screen.